

Introduction to Tropical Meteorology

First-Day Warm Up Activities

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Purpose

These activities are designed for senior undergraduates who have already completed a course in atmospheric dynamics. The goal of the first class is to connect familiar dynamical ideas to the tropical atmosphere and to introduce several questions that will motivate the rest of the course.

Activities: minutes

Activity	Duration
Activity 1: What is different about the tropics?	10 min
Activity 2: Does geostrophy work here?	10 min
Activity 3: Diagnose the tropical atmosphere from observations	10 min
Activity 4: Tropical meteorology mysteries	10 min

Opening Question: What Makes the Tropics Different?

Before discussion, answer the following individually:

What makes tropical atmospheric dynamics fundamentally different from midlatitude atmospheric dynamics?

Write down two or three ideas. Do not worry about whether they are complete.

Key Point: We will begin by comparing the physical environment of the tropics with the midlatitudes, then test whether familiar dynamical approximations remain valid.

Activity 1: What Is Dynamically Different About the Tropics?

Work in groups of 2-3.

You have learned atmospheric dynamics largely in a midlatitude context. Identify three to five assumptions, balances, or physical characteristics that may change as one approaches the equator.

For each item, discuss its likely dynamical consequence.

Possible categories to consider include:

- rotation, horizontal temperature gradients, moisture, convection, divergence and vertical motion, characteristic balances, waves and instabilities.

Record your ideas in the table below.

Quantity or Process	Midlatitudes	Tropics / Consequence
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Whole-class question: Which of these differences is most fundamental?

Key Point: One obvious difference is the weakening of the Coriolis parameter toward the equator. We can quantify the consequences using scale analysis.

Activity 2: Does Geostrophy Work Here?

For each system, calculate the Rossby number

$$Ro = \frac{U}{fL}.$$

System	U	L	f
A: Midlatitude cyclone	15 m s ⁻¹	1000 km	1.0×10^{-4} s ⁻¹
B: Tropical disturbance at 15°N	8 m s ⁻¹	500 km	3.8×10^{-5} s ⁻¹
C: Disturbance near 2°N	8 m s ⁻¹	1000 km	5.0×10^{-6} s ⁻¹

Answer the following:

1. Calculate Ro for each system.
2. Which system is most nearly geostrophic?
3. In which system is geostrophic balance most questionable?
4. If $f = 0$ at the equator, why does Earth's rotation still matter there?

Useful relation (here ϕ is the latitude)

$$f = 2\Omega \sin \phi.$$

Around any latitude $\phi = \phi_o$, we expand f into a series and approximate it as follows:

$$f = f_o + \beta y, \quad \beta = \left. \frac{\partial f}{\partial y} \right|_{\phi=\phi_o}$$

where $f_o = 2\Omega \sin \phi_o$. and y is the meridional distance from the reference latitude ϕ_o

Near the equator, consider the first-order approximation:

$$f \approx f_o + \beta y, \quad f \approx \beta y, \quad \beta = \left. \frac{\partial f}{\partial y} \right|_{\phi=0} = \frac{2\Omega}{a}.$$

where a is the radius of the earth.

Key Point: Scale analysis tells us that some familiar midlatitude balances weaken in the tropics. We now turn to observations and ask what a successful theory of the tropical atmosphere must actually explain.

Activity 3: Diagnose the Tropical Atmosphere from Observations

Next, I will display (also see last 2 pages of this document) several global or tropical maps, such as:

- sea-surface temperature, precipitation, omega

In groups, identify at least five major features that a theory of the tropical atmosphere should explain.

prediction: Before viewing the precipitation map, use the SST and low-level wind fields to predict where the greatest climatological rainfall should occur.

Discussion question:

Why does maximum tropical rainfall not simply coincide with maximum SST?

Key Point: The observations reveal organized structures that cannot be explained by temperature alone. Moisture, circulation, convection, and the ocean interact. We next turn these observations into a set of physical puzzles.

Activity 4: Tropical Meteorology Mysteries

Pick at least one question. Develop a plausible physical explanation using concepts you already know. You are not expected to give a complete answer.

1. Why are tropical cyclones rare within about 5° of the equator?
2. Why are tropical upper-tropospheric temperatures horizontally rather uniform?
3. Why are easterly winds common in the tropical lower troposphere?
4. Why is the ITCZ often displaced away from the equator?
5. How can waves exist and organize convection in an atmosphere with weak horizontal temperature gradients?
6. Why do warm oceans favor tropical cyclones, yet most regions with sufficiently warm SST do not produce them?

Prepare a one-minute explanation containing:

- your proposed mechanism,
- one dynamical or thermodynamic principle involved,
- one uncertainty or missing piece.

Key Point: These puzzles will reappear throughout the course.

Conclusion: We can now organize the course around the physical processes required to explain the features and puzzles identified today.

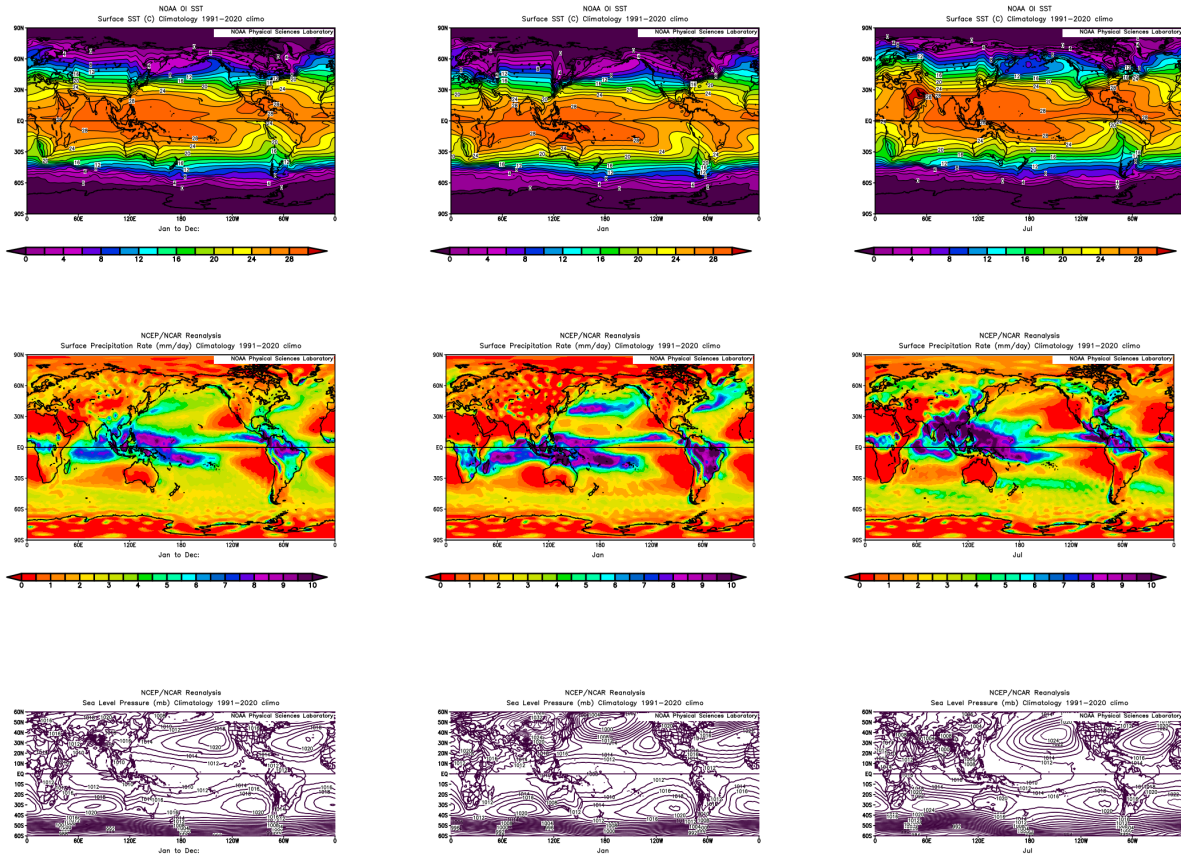


Figure 1: Long Term (1991-2020) mean fields. Source: NOAA PSD

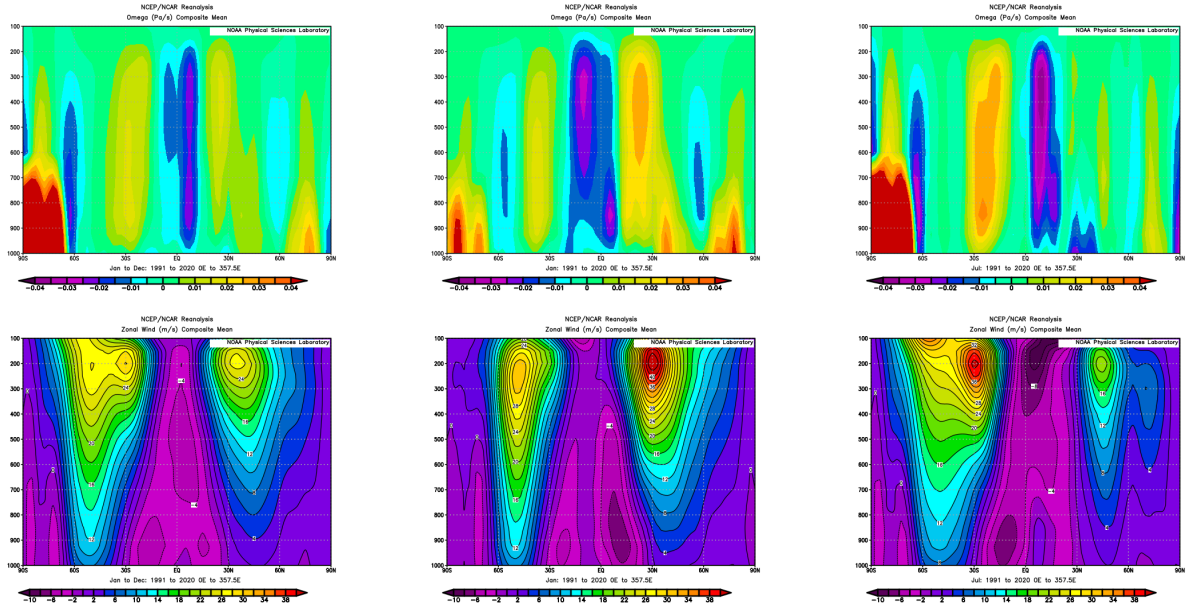


Figure 2: Long Term (1991-2020) zonally averaged mean fields (latitude-height section). Top panels show omega, and bottom panels show zonal winds. Source: NOAA PSD